

Unlearning a group from an in-context prompt: how much does corrupting it actually remove?

Abstract

We ask whether a group of training examples can be unlearned from a linear-attention in-context learner by *corrupting that group inside the prompt*, with no weight update, and whether a membership adversary is actually defeated or merely blinded. Sweeping six corruption families over four decades of strength, on a synthetic Gaussian task and on real MNIST features, across nine seed combinations, we find that **the membership signal lives in the first moment of the residual**. Additive-noise corruptions cannot reach it: they drive the adversary toward chance only by inflating variance, and a shared-noise control attributes 84–100% of that movement to masking rather than removal. Reaching chance this way costs $762\text{--}1699\times$ in preservation error on the synthetic task and $4375\text{--}7838\times$ on MNIST. Corruptions that move the *mean* — label flips — reach chance $6\times$ (synthetic) to $16\text{--}21\times$ (MNIST) more cheaply. We derive the flip’s effect on the read-out in closed form, exactly, which makes the membership AUC a differentiable function of the flip probability and yields $\theta^* = (N + 1)\delta_0/2 \sum_i a_i$ for the optimal flip. Optimising it lands on chance at $\text{AUC} = 0.4987 \pm 0.0005$ on MNIST, reproducibly across every seed, at $\varepsilon \approx 0.05$; the closed form agrees with the brief’s REINFORCE estimator and is $7\text{--}9\times$ faster. Where the required flip fraction exceeds one — as it does on one synthetic architecture — no label flip can cancel the gap even in principle, and we show the optimiser saturating against that boundary rather than converging.

1 Problem

Thesis. In this model the membership signal lives in the *first moment* of the residual. A corruption therefore erases it only if it moves the mean; every variance-based corruption can at best *mask* a signal it cannot reach, and pays for the masking in preservation error. The optimal corruption is the one that cancels the baseline mean gap — which we derive in closed form, and which correctly predicts the learned flip probability wherever the required cancellation is achievable.

A model is shown a prompt containing examples from three groups and must predict the query. One group is to be *forgotten*. We never touch the weights — we corrupt that group’s tokens in the prompt and ask whether a membership adversary can still tell that the group was trained on.

The eight results below are evidence for the boxed claim rather than a list of things that did or did not work. The negative results are not failures of particular corruptions; they are a map of where the signal is not.

Two quantities are in tension throughout.

Removal. An adversary sees the sign-aligned residual $r = \text{sign}(y)(\hat{y} - y)$ at a fixed probe point and must decide which of two models produced it: M_{full} , trained on all three groups, or M_{orac} , retrained without the forget group. We report the Mann–Whitney statistic

$$\text{AUC} = \Pr [r(M_{\text{full}}) > r(M_{\text{orac}})] + \frac{1}{2} \Pr [r(M_{\text{full}}) = r(M_{\text{orac}})], \quad (1)$$

so $\text{AUC} = \frac{1}{2}$ means the adversary is guessing. *Both models are scored on the same corrupted prompt with the same noise draw*, so detecting the edit cannot be mistaken for detecting membership.

The score is oriented by the usual membership-inference convention: the *member* hypothesis scores higher, so a working attack reads $\text{AUC} > \frac{1}{2}$ and chance is $\frac{1}{2}$. Since the full model has seen the forget group it fits forget-group queries *better* and its sign-aligned residual is smaller, the membership

score is the negated residual, $-r$. The ordering is fixed in the code rather than chosen per run, so it cannot be tuned after the fact. We do not symmetrise: $\max(a, 1 - a)$ is biased upward under the null (it reads ≈ 0.54 when the truth is 0.50), which would manufacture signal at exactly the point these experiments measure.

Preservation. How far the corrupted model has drifted from the retrain oracle it is meant to imitate:

$$\varepsilon = \text{KL}(p_{\text{oracle}} \parallel p_{\text{unlearned}}) \quad (2)$$

on the forget-population residual law, which is fit as a univariate Gaussian (μ, v) over the probe population, so ε is a Gaussian KL. Unlearning is only useful if $\text{AUC} \rightarrow \frac{1}{2}$ and ε stays small.

Are AUC and ε the same measurement? Both are functions of the first two moments, so it is fair to ask whether the cost ratios below are tautological. They are not, and the reason is the mechanism this report is about. AUC depends only on the *standardised* gap $(\mu_1 - \mu_0)/\sqrt{v_1 + v_0}$ and is invariant to rescaling both populations together; ε is not scale-invariant and grows with the variance ratio. A corruption that inflates v therefore drives AUC to $\frac{1}{2}$ while ε keeps climbing — which is exactly the masking regime, and exactly why the two can be traded against each other at all.

Corruptions. Applied to the forget tokens (x_f, y_f) only:

name	edit	parameter	channel
C1	$y_f \rightarrow y_f + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$	σ^2	variance
C2	$x_f \rightarrow x_f + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2 I)$	σ^2	variance
C3	both, independent draws	σ^2	variance
flip	$y_f \rightarrow (1 - 2t)y_f$	t	mean
bern	$y_f \rightarrow (1 - 2B)y_f, B \sim \text{Bern}(\theta)$	θ	mean + variance
whiten	$x_f \rightarrow$ covariance-matched to retain groups	mixing	—

The last two matter most. **flip** carries a mean shift with *no* variance; **bern** has the same mean, $\mathbb{E}[(1 - 2B)y] = (1 - 2\theta)y$, plus variance $4\theta(1 - \theta)y^2$. Writing the predicted AUC as $\Phi((\mu_1 - \mu_0)/\sqrt{v_1 + v_0})$ makes the difference concrete: a zero-mean corruption can only inflate the denominator, so it can never push AUC through $\frac{1}{2}$, while a mean shift moves the numerator directly.

2 Method

Algorithm 1 — measuring removal and preservation

Input: group spec, forget group f , corruption mode and grid

1. Train two shadow ensembles of $S = 512$ linear-attention models: M_{full} on all groups, M_{orac} on the retain groups only.
2. Draw a probe of $P = 64$ prompts and *freeze* it.
3. For each corruption strength t in the grid:
 - (a) corrupt the forget tokens of the probe $\rightarrow \tilde{X}$;
 - (b) run *both* ensembles on the same $\tilde{X}$;
 - (c) AUC(t) from the two residual populations;
 - (d) $\varepsilon(t)$ from the fitted residual laws;
 - (e) *shared-noise control*: repeat (a)–(c) broadcasting one noise draw across all S shadows. This holds variance inflation fixed, so the gap between the two runs separates *masking* from genuine removal.
4. Repeat for 3×3 (training seed, probe seed) combinations; report medians.

Nothing retrains inside the loop: the corruption is a prompt-time edit, so a whole sweep costs seconds against cached ensembles.

A closed form for the flip. The brief derives a REINFORCE estimator for Stage 2 because AUC is not differentiable through the sampling. For this policy that is avoidable. With $a_i = y_i (c_x \cdot x_i)$ over the forget slice, the induced change in the read-out has moments

$$\mathbb{E}[\Delta \hat{y}] = \frac{-2\theta}{N+1} \sum_i a_i, \quad \text{Var}[\Delta \hat{y}] = \frac{4\theta(1-\theta)}{(N+1)^2} \sum_i a_i^2, \quad (3)$$

exactly — because $(1 - 2B)^2 = 1$ leaves the label-label block of the context vector untouched. Feeding (3) into the Gaussian AUC makes $\text{AUC}(\theta)$ an ordinary differentiable function of θ . Monte Carlo at 4×10^5 trials per point confirms both moments to 3–4 significant figures.

Algorithm 2 — learning the corruption (Stage 2)

The brief writes $\min_{\theta} \text{AUC}(\theta) + \lambda \|\theta\|$. Every measured AUC here is *below* $\frac{1}{2}$, so minimising AUC drives toward *more* distinguishability once past $\frac{1}{2}$ — 0.30 is as good an adversary as 0.70, merely inverted. We minimise distance from chance instead:

$$\min_{\theta} \left(\text{AUC}(\theta) - \frac{1}{2} \right)^2 + \lambda \|\theta\|.$$

1. Initialise θ ; 2. $\text{AUC}(\theta)$ from (3) in closed form; 3. backprop and Adam step; 4. repeat 400 steps;
 5. report AUC and ε *measured* by Algorithm 1, not the closed-form value.
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Settings. Complete, for both datasets.

Data — synthetic		
input dimension	D	4
context length	N	31
groups		z_1, z_2, z_3 ; forget z_3
eigenbasis		identity, shared across groups
Λ_{z_1} eigenvalues	PR = 1.905	0.70, 0.15, 0.10, 0.05
Λ_{z_2} eigenvalues	PR = 2.381	0.60, 0.20, 0.10, 0.10
Λ_{z_3} eigenvalues	PR = 3.333	0.40, 0.30, 0.20, 0.10
covariances		trace-normalised, $\text{tr } \Lambda = 1$
labels		$y = \omega^\top x$, ω per sequence, shared across groups
data seed		0
Data — MNIST		
input source		MNIST images, PCA on pooled set to $D = 4$
groups		digits 1, 3, 8; forget digit 8
participation ratios		2.44, 3.29, 2.93
centring		per class; each bank trace-normalised
bank sizes		$\approx 6.7\text{k}$, 6.1k , 5.9k vectors
labels		$y = \text{sign}(\beta^\top x)$, β per sequence, shared
Probe (both)		
probe points	P	64
context token counts		10 / 10 / 11, forget group last
query		drawn from the forget group
probe seeds		777, 778, 779
Training (both)		
architectures		ATTN-S (factored), ATTN-M (merged)
shadow models per hypothesis	S	512
batch per shadow		8
steps		6000
optimiser		SGD, momentum 0.9
learning rate		0.005
gradient clipping		5.0, per shadow
initialisation scale		0.05
training seeds		1, 2, 3
Evaluation (both)		
seed combinations		$3 \times 3 = 9$; all values are medians
bootstrap resamples		200
shared-noise repeats		16
C1, C2 grid		21 points, $\sigma^2 \in [0, 100]$, log-spaced
C3 grid		9 points, $\sigma^2 \in [0, 10]$
flip grid		19 points, $t \in [0, 1.25]$
bern grid		21 points, $\theta \in [0, 1]$
whiten grid		9 points, mixture $\in [0, 1]$

Table 1: Complete experimental settings. Everything except the input source and the label rule is identical between the two datasets, so the comparison in Section 5 changes one thing at a time.

3 Control: does the attack measure anything?

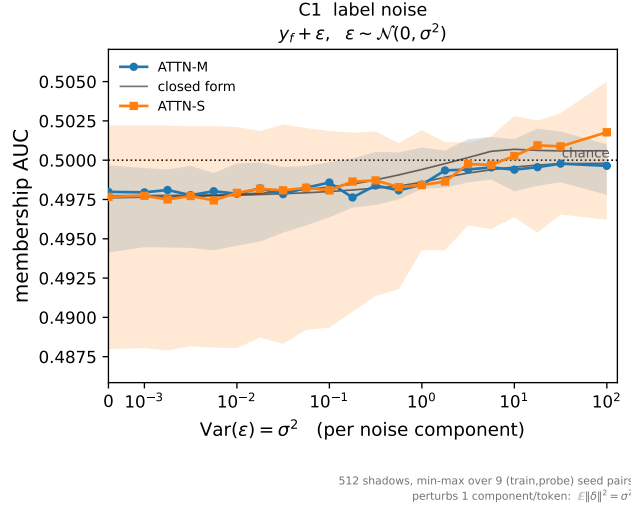


Figure 1: Null control — all three groups given the same covariance.

Give every group the same spectrum and the same eigenbasis. Then M_{full} and M_{orac} trained on the same distribution, there is genuinely no membership information, and the pipeline *must* return chance.

It does: median AUC = 0.498 for both architectures across the whole grid, spanning 0.4976–0.4998 and 0.4974–0.5018. The real experiments below sit at 0.535 and 0.625 — far outside that band. Whatever those numbers are, they are not an artifact of the attack code.

4 The setup, at a glance

Before the results, the whole apparatus on one page.

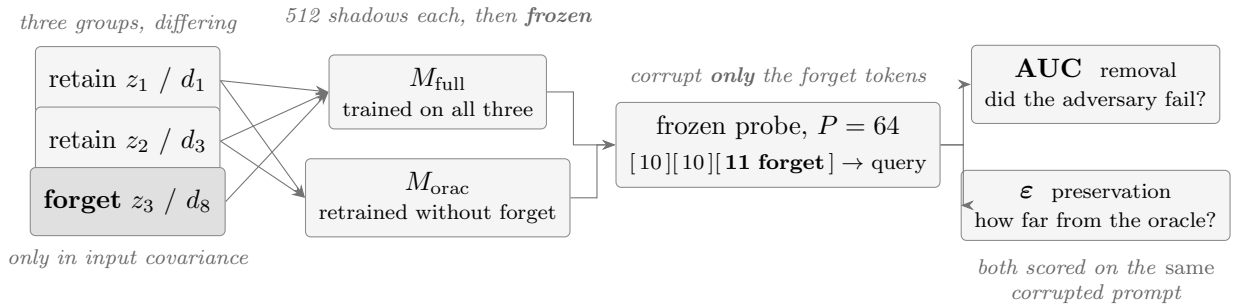


Figure 2: The pipeline. Groups differ only in the shape of their input covariance; all share one labelling rule. Two shadow ensembles are trained once and frozen — M_{full} has seen the forget group, M_{orac} has not. At evaluation the forget tokens of a frozen probe are corrupted and *both* ensembles are scored on that same corrupted prompt, so edit-detection cannot masquerade as membership. Each sweep point yields a removal number (AUC) and a preservation number (ε). Nothing retrains inside the loop.

Reading the numbers.

- AUC = $\frac{1}{2}$ is the *goal* — the adversary guessing. The baseline sits *above* $\frac{1}{2}$ (a working attack) and unlearning means driving it *down* to $\frac{1}{2}$. Overshooting below $\frac{1}{2}$ is not success either: the attack has merely inverted, and $|\text{AUC} - \frac{1}{2}|$ remains the signal.

- ε small is good. It is a KL against the retrain oracle, so $\varepsilon \rightarrow 0$ means “indistinguishable from having retrained properly”.
- **Cost ratio** = ε at the closest approach to chance, divided by ε at its own minimum. It is the price of defeating the adversary, and it is the number the whole report turns on.
- Every value is a median over nine (training seed, probe seed) combinations; bands are min-max across those nine.

5 Experiment 1 — hand-designed corruption (synthetic)

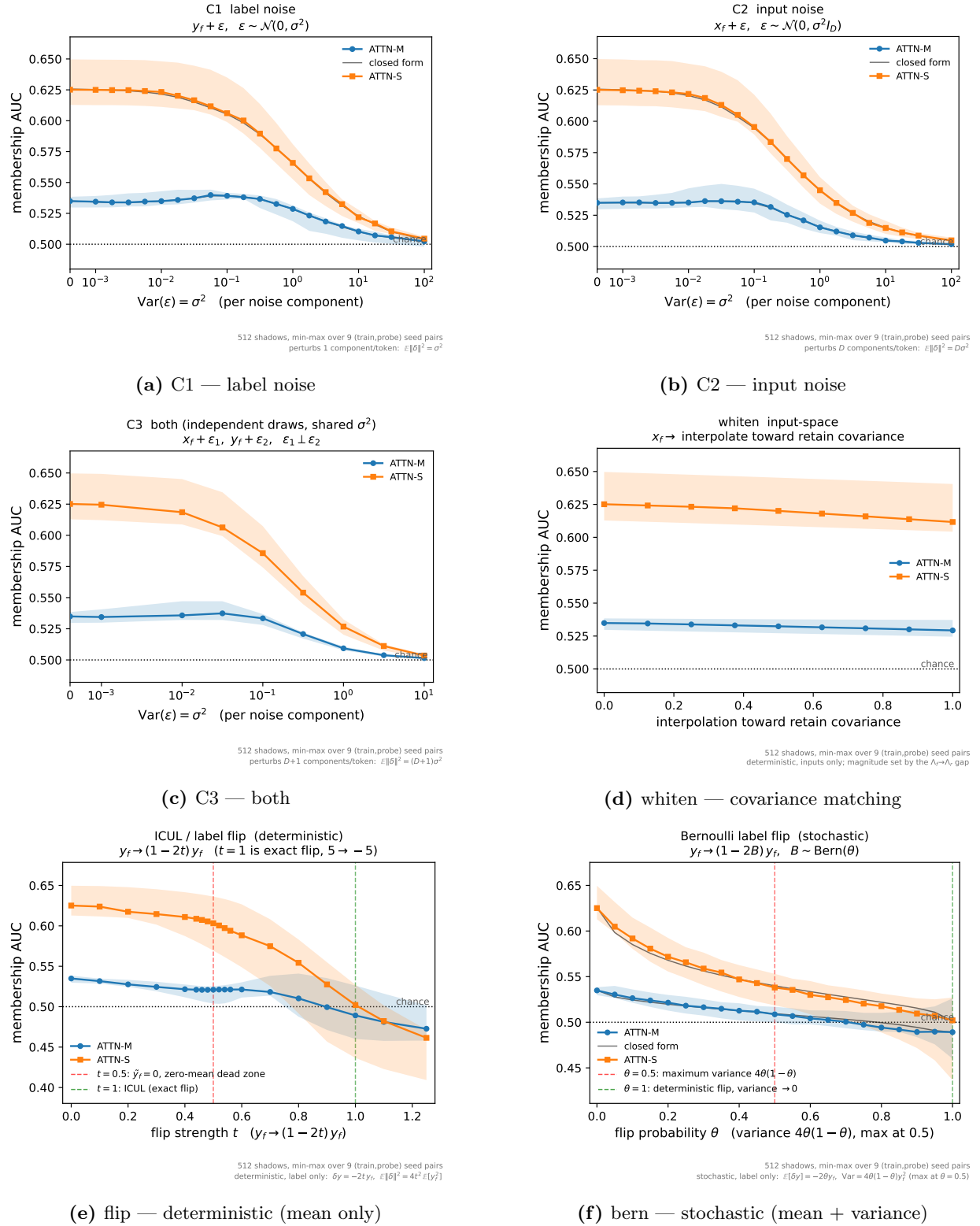
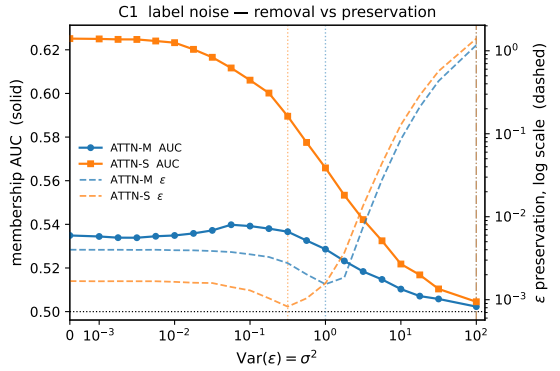


Figure 3: All six corruption arms, synthetic task. Membership AUC against corruption strength. Bands: min-max over nine seed combinations. Thin line: closed-form prediction. Dotted: chance. Note the bottom row: **flip** and **bern** both rise *through* chance, while none of the four arms above them ever reaches it.

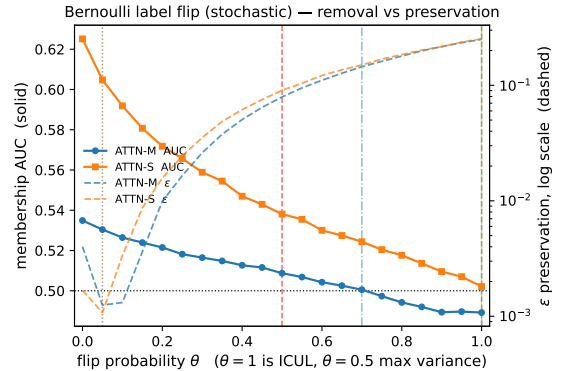
At zero corruption the adversary achieves $\text{AUC} = 0.535$ (ATTN-M) and 0.625 (ATTN-S); per-training-seed medians span only $0.5332\text{--}0.5352$ and $0.6223\text{--}0.6275$, so the baselines are systematic, not sampling noise. AUC falls monotonically toward chance, reaching 0.502 and 0.504 at $\sigma^2 = 100$. The closed form tracks the Gaussian arms to 6.4×10^{-4} mean absolute error (max 2.9×10^{-3}); the Bernoulli arm is looser (3.2×10^{-3} mean, 1.7×10^{-2} max) because its mean shift sums only $n_f = 11$ Bernoulli terms and is visibly non-Gaussian near $\theta = 1$.

arm	arch	best preservation		closest to chance	
		$\varepsilon_{\min}$	at	AUC (ε)	cost ratio
C1	ATTN-M	0.00154	$\sigma^2 = 1.00$	0.5023 (1.170)	$762\times$
C1	ATTN-S	0.00082	$\sigma^2 = 0.32$	0.5045 (1.402)	$1699\times$
C2	ATTN-M	0.00116	$\sigma^2 = 0.18$	0.5019 (1.347)	$1165\times$
C2	ATTN-S	0.00088	$\sigma^2 = 0.10$	0.5049 (1.435)	$1630\times$
C3	ATTN-M	0.00158	$\sigma^2 = 0.10$	0.5015 (1.344)	$848\times$
C3	ATTN-S	0.00102	$\sigma^2 = 0.10$	0.5035 (1.405)	$1379\times$
flip	ATTN-M	0.00157	$t = 0.10$	0.4994 (0.209)	$133\times$
flip	ATTN-S	0.00139	$t = 0.10$	0.5021 (0.252)	$181\times$
bern	ATTN-M	0.00125	$\theta = 0.05$	0.5005 (0.145)	$116\times$
bern	ATTN-S	0.00108	$\theta = 0.05$	0.5021 (0.252)	$234\times$
whiten	ATTN-M	0.00134	$m = 0.75$	0.5293 (0.002)	never reaches
whiten	ATTN-S	0.00133	$m = 0.25$	0.6117 (0.004)	never reaches

Table 2: All six arms on identical data. “Cost ratio” is ε at the closest approach to chance divided by $\varepsilon_{\min}$ — the preservation bill for defeating the adversary. The “at” column gives the corruption strength at which ε bottoms out, *in each arm’s own parameter*: noise variance σ^2 , deterministic flip strength t , flip probability θ , or whitening mixture m . Those are not on a common scale and should not be compared across blocks.

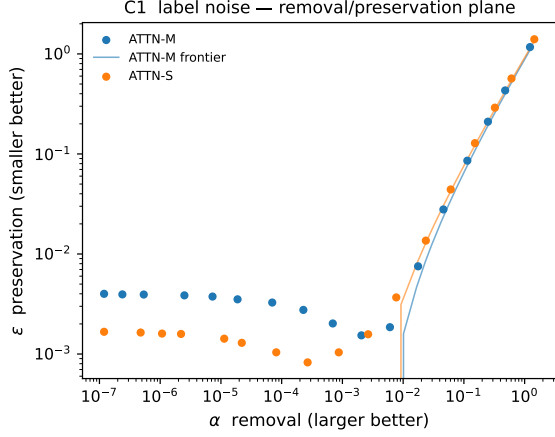


(a) C1 — AUC and ε against σ^2

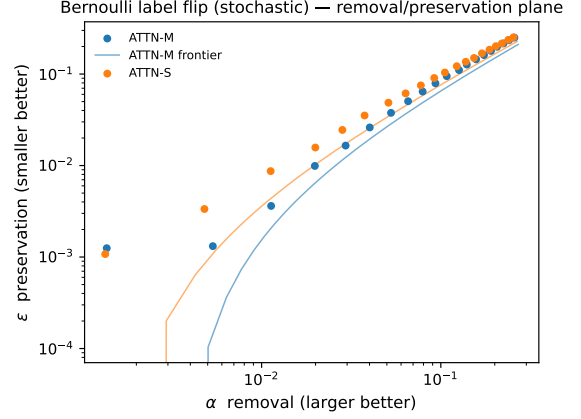


(b) bern — AUC and ε against θ

Figure 4: The tradeoff, directly. AUC (solid, left axis) rises toward chance while ε (dashed, right axis, log scale) climbs three decades. Vertical markers: the ε minimum and the closest approach to chance. The gap between those two markers *is* the cost ratio in Table 2 — and it is visibly narrower for **bern**.



(a) C1 — Gaussian noise



(b) bern — stochastic flip

Figure 5: The removal/preservation plane. Each point is one corruption strength: removal α rightward, preservation error ϵ upward. A usable unlearning method would reach the bottom-right. Both arms instead turn sharply upward — past a threshold, further removal is bought almost entirely with preservation error.

Result 1 (negative). No arm offers a clean operating point. Reaching chance costs between two and three orders of magnitude in preservation error. Figure 5 shows why in one picture: the frontier is flat while removal is cheap, then turns almost vertical.

Result 2. Most of the apparent progress is *masking*, not removal. The shared-noise control attributes 100%, 84% and 87% of the movement at $\sigma^2 = 0.32, 0.56, 1$ to variance inflation hiding the signal, falling to 41% only at $\sigma^2 = 100$. (Below $\sigma^2 \approx 0.1$ the decomposition is undefined — AUC does not move toward chance there at all — and no claim here rests on that region.)

Result 3. The arms are not equally bad, and they separate by *which moment* they move. The Gaussian arms never go below 0.501; the deterministic flip falls *through* chance to 0.473 and 0.461, exactly as the Φ argument above predicts. Whitening barely moves at all (0.535 $\rightarrow$ 0.529). The stochastic flip, carrying both channels, is the cheapest arm in the table by a factor of ~ 7 .

6 Experiment 2 — replication on real data

How MNIST enters the experiment

MNIST supplies the *input distributions*, not the task. This is not digit classification, and the construction is worth stating precisely because it is easy to assume otherwise.

Three groups. Take digits 1, 3 and 8; PCA the *pooled* set down to $D = 4$ so all three share one feature space; split back by digit; centre each class and normalise it so $\text{tr}(\text{Cov}) = 1$. The result is three *banks* of 4-dimensional vectors — roughly 6.7k, 6.1k and 5.9k rows, one per image. A group is therefore a cloud of thousands of points, and because the traces are equal the clouds differ only in covariance *shape*, summarised by the participation ratio $\text{PR} = (\text{tr } \Lambda)^2 / \text{tr}(\Lambda^2)$: 2.44, 3.29, 2.93. **Retain** = $\{d_1, d_3\}$, **forget** = d_8 . Note the forget group is the middle one, not an extreme — unlike the synthetic setup, the geometry here is whatever the data happened to provide.

The labels come from a random plane, not from the digit. For each prompt we draw a fresh $\beta \sim \mathcal{N}(0, I_4)$ and set $y = \text{sign}(\beta^\top x)$. The label alphabet is fixed at ± 1 , but the labelling

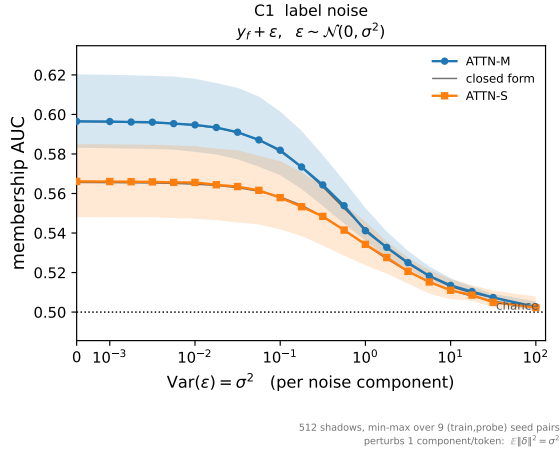
function changes every prompt, so the same image receives different labels in different prompts. A fixed image→digit rule would be memorisable in the weights, leaving nothing to learn in context; a fresh β forces the model to recover the plane from the examples in front of it.

β is shared across the three groups. This is the load-bearing constraint. If each digit carried its own labelling rule, corrupting the forget group would delete the answer rather than the membership evidence, and a fall in AUC would measure nothing. With β shared, all three groups obey one rule and differ *only* in input geometry — which is precisely the quantity the attack is being asked to detect. It also keeps the retrain oracle competent on forget-group queries: it can label a d_8 point perfectly well, having simply never seen that class’s geometry during training. Had the digit been the label, the oracle could not answer at all and membership inference would collapse into trivial capability detection.

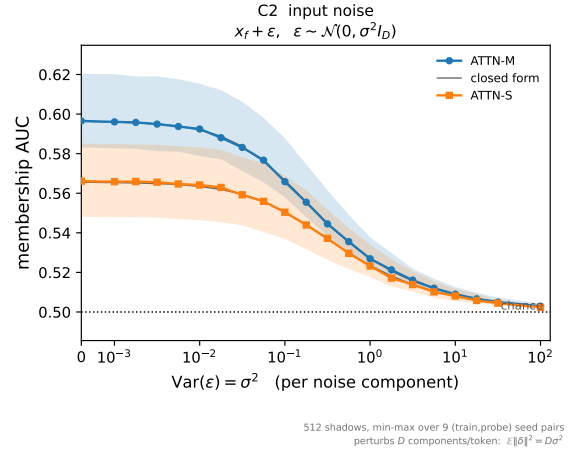
Probe layout. Each of the $P = 64$ prompts is

$$\underbrace{[10 \text{ from } d_1][10 \text{ from } d_3][11 \text{ from } d_8]}_{N=31 \text{ context tokens}} \longrightarrow \underbrace{1 \text{ more } d_8 \text{ point}}_{\text{query}}$$

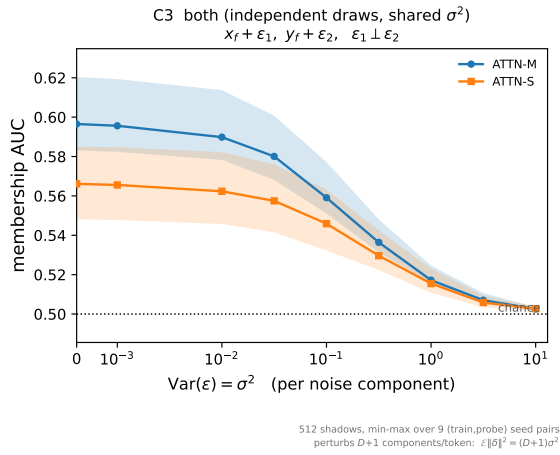
Forget tokens are placed last so they form a contiguous slice, and the *query is drawn from the forget group* — the model is asked about the distribution we are trying to make it forget, which is where the membership signal appears. Every corruption arm edits only those 11 d_8 context tokens: never d_1 , never d_3 , never the query. Everything else — the two 512-model ensembles, the attack, the sweep, the shared-noise control — is unchanged from Experiment 1.



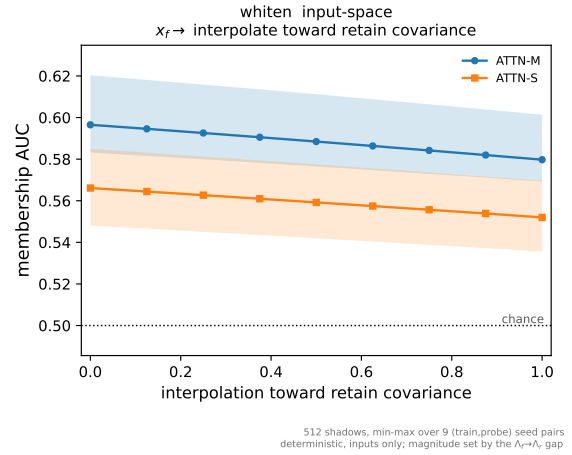
(a) C1 — label noise



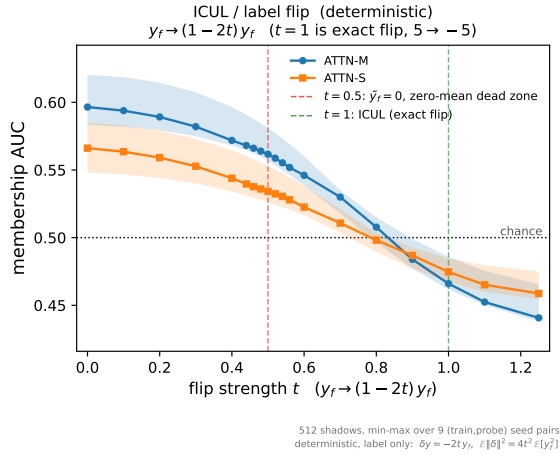
(b) C2 — input noise



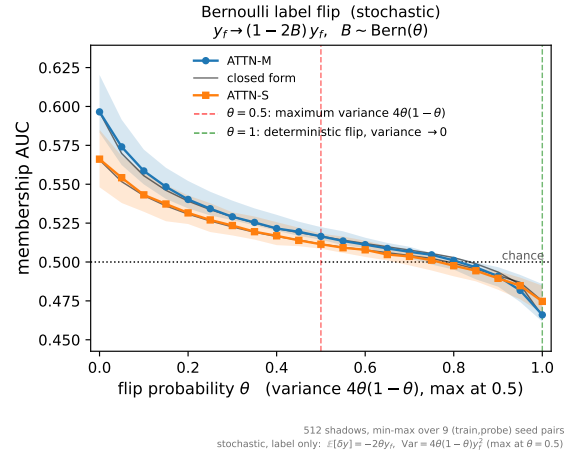
(c) C3 — both



(d) whiten — covariance matching

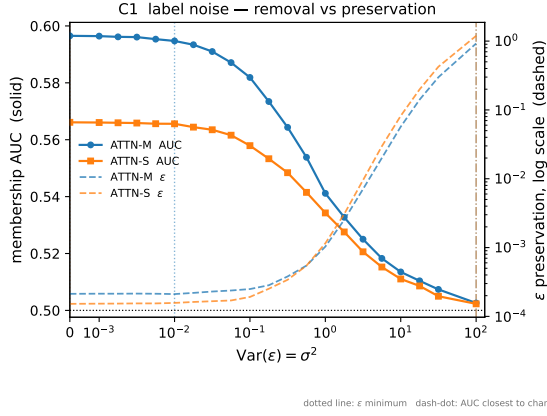


(e) flip — deterministic

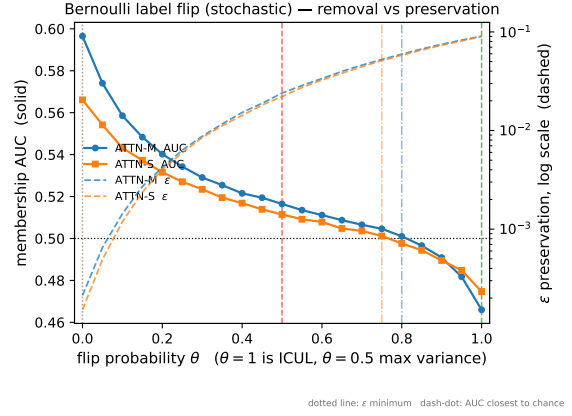


(f) bern — stochastic

Figure 6: All six arms on MNIST (digits 1/3/8, forget digit 8). Same pipeline, real image features. The baselines start lower — a stronger membership signal — and the same ordering of arms holds.



(a) C1 — Gaussian noise



(b) bern — stochastic flip

Figure 7: MNIST tradeoff. The same pair of markers as Figure 4, on real data. The separation between them is far wider for the Gaussian arm here than it was synthetically — this is Result 4.

Result

The synthetic task fixes the input distribution by construction, which is what makes the closed form checkable but also what makes the setup artificial. The question here is whether the conclusion survives when the input clouds are real and messy — genuinely non-Gaussian, correlated, whatever handwritten digits happen to be — rather than designed.

It does, and the gap widens. Baseline AUC is 0.596 (ATTN-M) and 0.566 (ATTN-S), a *stronger* membership signal than the synthetic task, and the closed form is if anything more accurate here (4.6×10^{-4} and 3.8×10^{-4} mean absolute error on C1 and C2).

arm	arch	best preservation		closest to chance	
		$\varepsilon_{\min}$	at	AUC (ε)	cost ratio
C1	ATTN-M	0.00021	$\sigma^2=0.01$	0.5026 (0.923)	4375×
C1	ATTN-S	0.00015	$\sigma^2=0$	0.5023 (1.193)	7838×
C2	ATTN-M	0.00021	$\sigma^2=0.001$	0.5029 (1.020)	4798×
C2	ATTN-S	0.00015	$\sigma^2=0$	0.5023 (1.084)	7122×
bern	ATTN-M	0.00021	$\theta=0$	0.5010 (0.059)	279 ×
bern	ATTN-S	0.00015	$\theta=0$	0.5012 (0.051)	337 ×

Table 3: MNIST. Columns as Table 2; again the “at” values are in each arm’s own parameter. Here every minimum sits at or beside zero corruption.

Result 4. The negative result replicates and is *worse* on real data: 4375–7838× against 762–1699× synthetically.

Result 5. The flip’s advantage replicates but its magnitude does not transfer: 5.9–6.6× cheaper than the best Gaussian arm on the synthetic task, 15.7–21.1× on MNIST. The direction is robust; the number should never be quoted without naming the dataset.

What does *not* explain Result 5. The obvious hypothesis is that the advantage tracks the baseline mean gap — a larger gap needs a larger flip, which costs more. It does not. Writing the advantage as $\varepsilon_{\text{gauss}}/\varepsilon_{\text{bern}}$ at the closest approach to chance (the $\varepsilon_{\min}$ cancels), and the standardised baseline gap as $z_0 = \Phi^{-1}(\text{AUC}_0)$:

dataset	arch	z_0	$\varepsilon_{\text{gauss}} / \varepsilon_{\text{bern}}$	advantage
synthetic	ATTN-M	-0.088	1.170/0.145	8.1×
synthetic	ATTN-S	-0.319	1.402/0.252	5.6×
MNIST	ATTN-M	-0.244	0.923/0.059	15.5×
MNIST	ATTN-S	-0.167	1.084/0.051	21.1×

Across the four cases $|z_0|$ and the advantage correlate at -0.23 — no relationship. The driver is visible in the fourth column instead: $\varepsilon_{\text{gauss}}$ is comparable everywhere (0.92–1.40), while $\varepsilon_{\text{bern}}$ is three to five times smaller on MNIST than on the synthetic task. The flip is not reaching chance more easily on MNIST; it is reaching it *more cheaply*. A natural mechanism — consistent with these numbers but not tested here — is that a flip’s preservation cost scales with the label second moment, which is bounded for MNIST’s ± 1 labels and much larger for continuous regression targets. This is a boundary of the moment argument: it predicts *where* the optimum sits, not what it costs. One thing that does *not* replicate: synthetically, ε is minimised at non-zero corruption — a little corruption moves the model toward the oracle. On MNIST the minimum sits at $\sigma^2 \approx 0$ for every arm.

7 Experiment 3 — learning the corruption

Experiment 1 swept θ by hand. Here Algorithm 2 optimises it.

dataset	arch	closed form (Alg. 2)		REINFORCE	
		θ^*	AUC (ε)	θ	AUC
MNIST	ATTN-S	0.678	0.4993 (0.036)	0.703	0.4975
MNIST	ATTN-M	0.753	0.4982 (0.049)	0.737	0.4997
synthetic	ATTN-S	0.949	0.5260 (0.264)	0.622	0.5423
synthetic	ATTN-M	0.850	0.5106 (0.199)	0.016	0.5386

Table 4: Both gradient estimators. The closed form takes 1.7–2.0s against 14.1–14.5s for REINFORCE — 7–9× faster, and deterministic.

dataset	arch	θ^*	AUC achieved	ε
MNIST	ATTN-S	0.761 ± 0.061	0.4987 ± 0.0005	0.053 ± 0.012
MNIST	ATTN-M	0.816 ± 0.043	0.4984 ± 0.0005	0.061 ± 0.008
synthetic	ATTN-S	0.895 ± 0.091	0.5074 ± 0.0143	0.185 ± 0.065
synthetic	ATTN-M	0.739 ± 0.138	0.5021 ± 0.0064	0.139 ± 0.081

Table 5: Across all nine seed combinations, mean $\pm$ s.d.

Result 6 (positive). On MNIST the learned flip lands at $\text{AUC} = 0.4987 \pm 0.0005$ — chance, to within a twentieth of a percentage point, reproducibly across every seed — at $\varepsilon \approx 0.05$, against $\varepsilon \approx 1.0$ for Gaussian noise to reach the same place. Three independent routes agree: the closed form gives $\theta = 0.678/0.753$, a 41-point grid search gives $0.675/0.725$, and REINFORCE lands within 0.03.

Result 7 (limitation). It does not transfer to the synthetic task. There the two estimators diverge wildly — $\theta = 0.850$ versus 0.016 on ATTN-M — and neither reaches chance (0.5106 and 0.5386). The closed form’s Bernoulli error is 3.2×10^{-3} there against 1.1×10^{-3} on MNIST. Result 9 below gives the reason, and it is structural rather than numerical.

Result 8 — and a closed form for θ^* . The optimum flips 68–82% of forget labels: not a small perturbation, but the *mean-compensating* solution. Setting the corrupted mean gap to zero, $\delta_0 + \mathbb{E}[\Delta\hat{y}] = 0$ with $\mathbb{E}[\Delta\hat{y}]$ from (3), gives

$$\theta^* = \frac{(N+1)\delta_0}{2\sum_i a_i}, \quad (4)$$

where δ_0 is the baseline mean gap. Note what (4) does *not* contain: the variance. The extra spread $4\theta(1-\theta)\sum_i a_i^2$ sits in the denominator of the AUC and cannot move the point where the numerator vanishes.

That yields a prediction testable on data already in hand. The deterministic **flip** at strength t has the *same* mean shift and no variance at all, so its AUC must cross $\frac{1}{2}$ at exactly $t = \theta^*$. Reading that crossing off the **flip** sweep predicts the learned θ^* with no fitting:

dataset	arch	flip crosses $\frac{1}{2}$ at	bern crosses at	learned θ^*
MNIST	ATTN-S	0.785	0.767	0.761 ± 0.061
MNIST	ATTN-M	0.833	0.812	0.816 ± 0.043
synthetic	ATTN-M	0.895	0.708	0.739 ± 0.138
synthetic	ATTN-S	1.011	never	0.895 ± 0.091

Table 6: Predicted versus realised flip fraction. On MNIST the three columns agree to within 0.02 — the closed form predicts the learned policy. On the synthetic task they do not.

Result 9 — why the synthetic task fails, structurally. The last row of Table 6 is the explanation for Result 7. Cancelling the baseline gap on synthetic ATTN-S requires $t = 1.011$. But θ is a *probability*: $\theta \leq 1$. **The required mean cancellation lies outside the range a label flip can deliver**, so **bern** cannot reach chance there in principle — and indeed it never does, bottoming out at 0.5021. The optimiser returning $\theta^* = 0.895 \pm 0.091$ is not converging badly; it is saturating against a boundary. Synthetic ATTN-M sits just inside the feasible region and is correspondingly noisy.

This also explains the estimator disagreement quantitatively. Pushing θ^* toward 1 drives the flip’s mean shift — a sum of $n_f = 11$ Bernoulli terms — into its degenerate corner, where the variance vanishes and the standardised third moment diverges. At the fitted optima the skew of that sum is -0.37 and -0.49 on MNIST against -0.33 and -0.78 on synthetic, and on the worst synthetic seed ($\theta = 0.964$) it reaches -1.50 with excess kurtosis $+2.07$. The Gaussian AUC underlying (3) is therefore weakest exactly where the synthetic optimum is forced to sit. (These are equal-weight values; unequal a_i inflate them further, and the synthetic task’s continuous labels give a much wider spread of a_i than MNIST’s ± 1 .)

8 Summary

#	finding	evidence
<i>Where the signal is not — variance cannot reach it</i>		
1	No corruption family reaches chance cheaply	116–7838× in ϵ
2	Most movement toward chance is masking, not removal	shared-noise control, 84–100%
3	Arms separate by which moment they move	only mean-shifting arms cross $\frac{1}{2}$
4	The negative result is worse on real data	4375–7838× on MNIST
<i>Where it is — the first moment, and it can be computed</i>		
8	The optimum is mean-compensating and is predicted in closed form by (4)	flips 68–82% of labels agrees to 0.02 on MNIST
6	A learned flip does reach chance, on MNIST	0.4987 ± 0.0005 , $\epsilon \approx 0.05$
<i>Where the argument stops</i>		
9	Synthetic ATTN-S needs $\theta^* > 1$ — infeasible	flip crosses at $t = 1.011$
7	...so Stage 2 saturates and the estimators diverge	skew -1.50 at the worst seed
5	Cost, unlike position, is not predicted by the gap	$\text{corr}(z_0 , \text{adv}) = -0.23$

In one sentence. Context-level unlearning cannot remove group membership, only mask it — unless the corruption moves the exact moment the signal lives in, which for this model is the mean; the optimal such corruption is the one cancelling the baseline gap, it is available in closed form, and it works wherever that cancellation is achievable at all.

Scope. This is *group-level* unlearning: the adversary detects whether a distribution was trained on, not whether a specific example was. Two datasets, one architecture family, scalar policy only — the conditional policy $p_\theta(\cdot | x_f, y_f)$ is implemented but not reported here. Fixing Result 7 by polishing the closed-form optimum against measured AUC is the obvious next step.

Two bugs found and fixed. Stage 2’s measurement originally treated the flip probability θ as if it were a realised draw, computing $(1 - 2\theta)y_f$ — which is the deterministic **flip** arm, not the stochastic one. It was caught by cross-checking against Experiment 1: the Stage 2 grid search reproduced the **flip** curve exactly (0.4751) instead of the **bern** curve (0.4821). Separately, the sweep script hardcoded an eigenbasis field that MNIST specs do not have, so the MNIST sweep crashed on bookkeeping after computing every result. All numbers above are post-fix.

A Derivation of the flip moments and of θ^*

Equation (3) is stated in Section 2 without proof. It is four lines.

Step 1 — the read-out is linear in the context labels. For linear attention performing in-context regression, the prediction at the query is a weighted sum of the context labels,

$$\hat{y}_q = \frac{1}{N+1} \sum_i y_i (c_x \cdot x_i), \quad c_x = M^\top x_q.$$

Write $a_i = y_i (c_x \cdot x_i)$ for the signed contribution of context token i , so that $\hat{y}_q = \frac{1}{N+1} \sum_i a_i$. Linearity in y is the only property used below.

Step 2 — what the flip does. The policy replaces y_i by $(1 - 2B_i)y_i$ for i in the forget slice, with $B_i \sim \text{Bern}(\theta)$ drawn independently. Token i 's contribution becomes $(1 - 2B_i)a_i$, so the change in the prediction is

$$\Delta \hat{y} = \frac{1}{N+1} \sum_{i \in f} [(1 - 2B_i) - 1] a_i = \frac{-2}{N+1} \sum_{i \in f} B_i a_i.$$

A weighted sum of independent Bernoulli variables.

Step 3 — its moments. With $\mathbb{E}[B_i] = \theta$ and $\text{Var}(B_i) = \theta(1 - \theta)$, independence gives (3) immediately:

$$\mathbb{E}[\Delta \hat{y}] = \frac{-2\theta}{N+1} \sum_i a_i, \quad \text{Var}[\Delta \hat{y}] = \frac{4\theta(1 - \theta)}{(N+1)^2} \sum_i a_i^2.$$

Step 4 — why this is exact, not first-order. The context vector also carries a label-label block, terms in $\sum_i y_i^2$. Under the flip those become $\sum_i (1 - 2B_i)^2 y_i^2 = \sum_i y_i^2$, because $(1 - 2B)^2 = 1$ for $B \in \{0, 1\}$: the block is *unchanged*. No second-order correction survives, so the moments hold exactly for every θ . Contrast additive noise, $y \rightarrow y + \epsilon$, which sends $\sum y^2 \rightarrow \sum y^2 + 2 \sum y\epsilon + \sum \epsilon^2$ and leaves terms behind.

Consequence 1 — a differentiable AUC. Modelling the two residual populations as Gaussian, $\text{AUC} = \Phi((\mu_1 - \mu_0)/\sqrt{v_1 + v_0})$, and (3) supplies $\mu_1(\theta)$ and $v_1(\theta)$ in closed form. $\text{AUC}(\theta)$ is therefore an explicit smooth function and can be differentiated directly — no sampling, no gradient variance. This is Algorithm 2.

Consequence 2 — the optimum. Setting the corrupted mean gap to zero, $\delta_0 + \mathbb{E}[\Delta \hat{y}] = 0$, and solving gives (4), $\theta^* = (N+1)\delta_0/(2 \sum_i a_i)$. The variance term does not appear: it sits in the denominator of the AUC and cannot move the point at which the numerator vanishes. That is why the deterministic flip — same mean shift, no variance — must cross $\frac{1}{2}$ at the same parameter value, which is the prediction tested in Table 6.

Verification. `tests/verify_bern.py` draws 4×10^5 Monte Carlo realisations per grid point and recovers both moments to 3–4 significant figures, and confirms $\sum \tilde{y}^2 - \sum y^2 = 0$ exactly.

The two estimators, side by side

Closed form (Algorithm 2)	REINFORCE (the brief)
Uses the formula for $AUC(\theta)$ above and differentiates it.	Never uses the formula. Samples masks and infers the slope from measurements.
Per step: evaluate a formula, autograd, Adam step. No model is run.	Per step: draw 32 masks, apply each, run both 512-model ensembles, weight by the score $\nabla_{\theta} \log p_{\theta}(B) = \sum_i [B_i/\theta - (1 - B_i)/(1 - \theta)]$.
Deterministic — identical answer every run.	Stochastic — a different answer every run; unbiased but high variance.
1.7–2.0 s.	14.1–14.5 s.
Requires the read-out to be linear in the labels. Lost if the architecture changes.	Requires only the ability to sample a corruption and measure the result. Works for any policy, any model, differentiable or not.

The brief’s derivation is correct and general; it is simply unnecessary for *this* policy, because the model’s linearity in the labels lets the expectation be computed rather than sampled. Running both is a check on the derivation: they optimise the same objective by independent routes, so agreement on MNIST (within 0.03 in θ) is evidence that (3) is right.

B Preservation error against corruption strength

Figures 3 and 6 plot removal. These plot the other side of the ledger: how ε grows as the corruption is turned up. The minimum of each curve is the $\varepsilon_{\min}$ column of Tables 2 and 3.

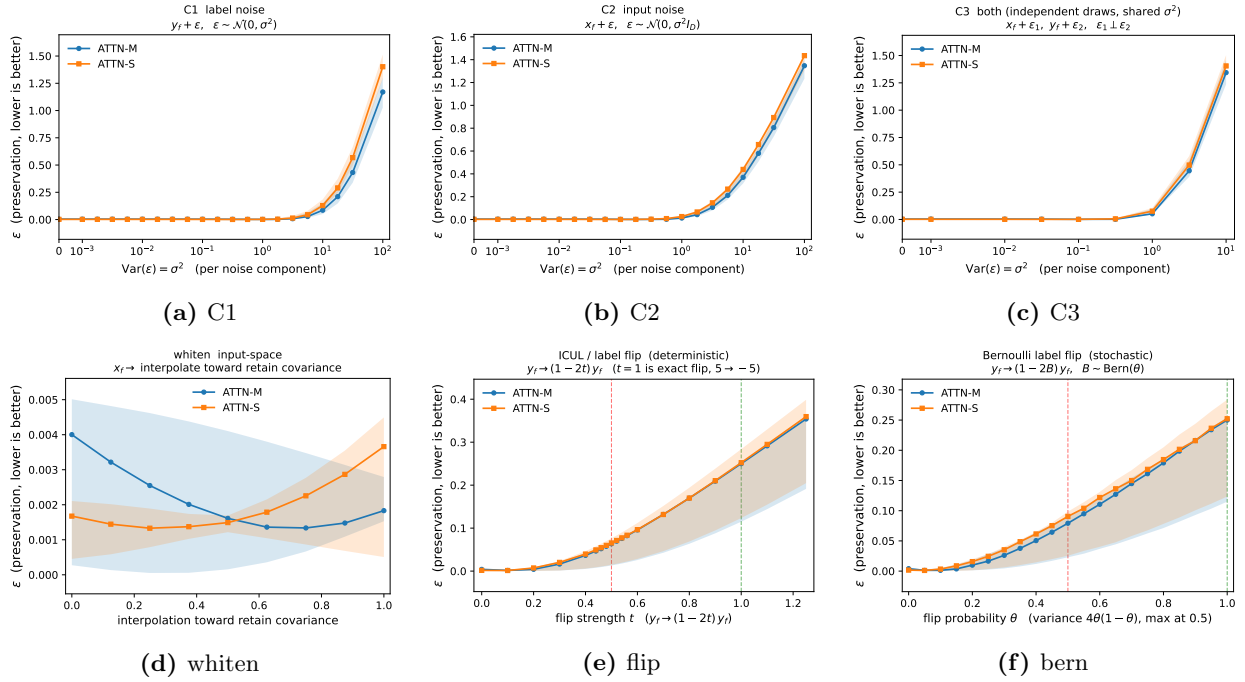


Figure 8: Preservation error, synthetic task. Note the dip below the zero-corruption value in several arms — mild corruption moves the unlearned model *toward* the oracle before the cost takes over.

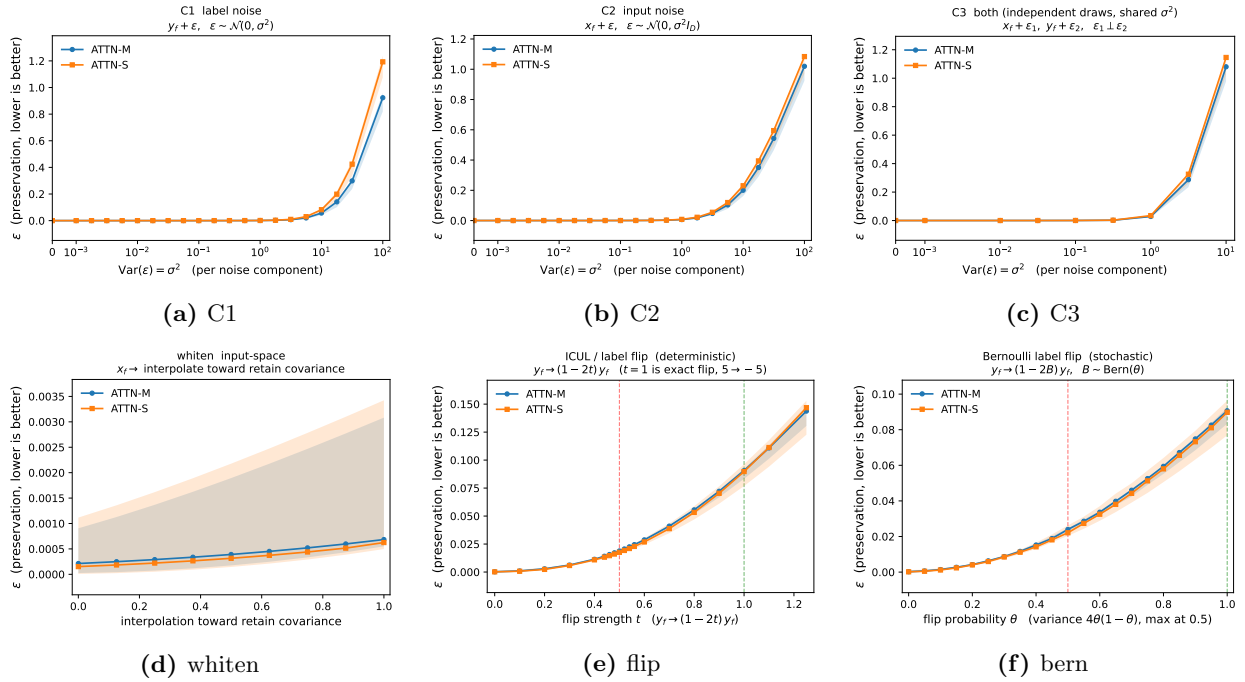


Figure 9: Preservation error, MNIST. The dip is absent here — the minimum sits at zero corruption for every arm.

C Removal/preservation frontiers, all arms

Each point is one corruption strength, plotted as removal α (rightward, larger better) against preservation error ε (upward, smaller better). A method with a clean operating point would produce a frontier that extends to the right while staying low. None does.

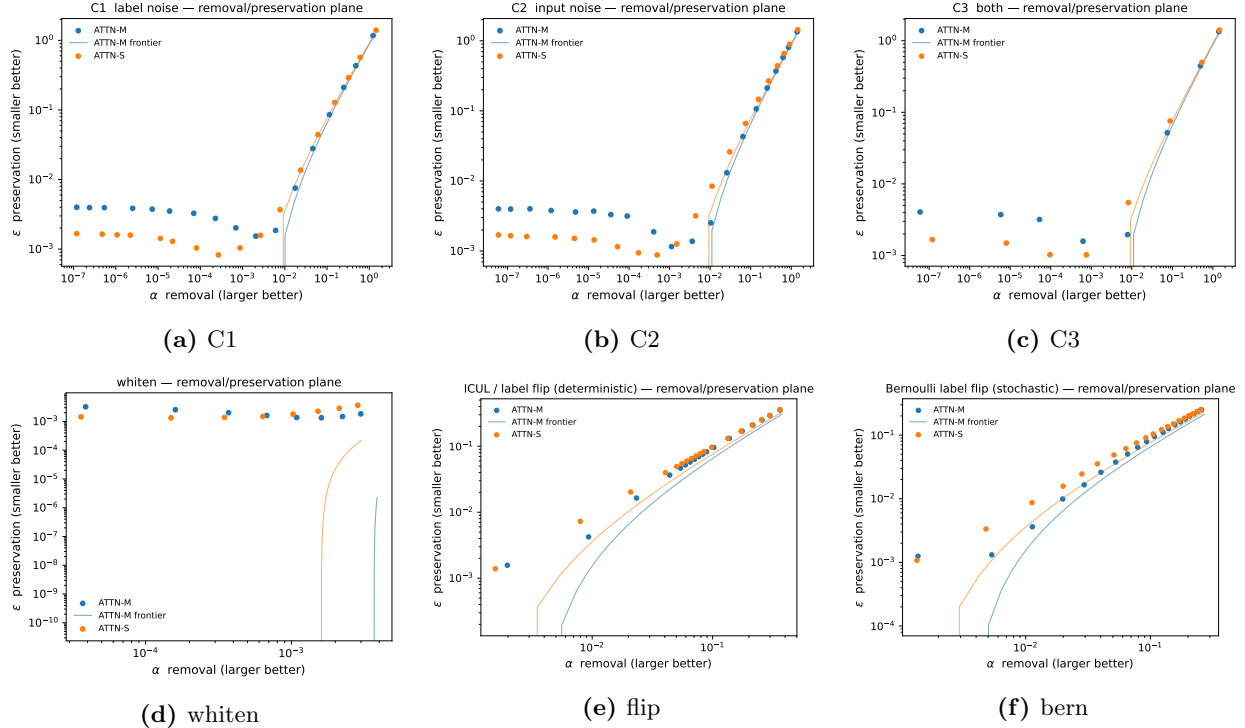


Figure 10: Frontiers, synthetic task.

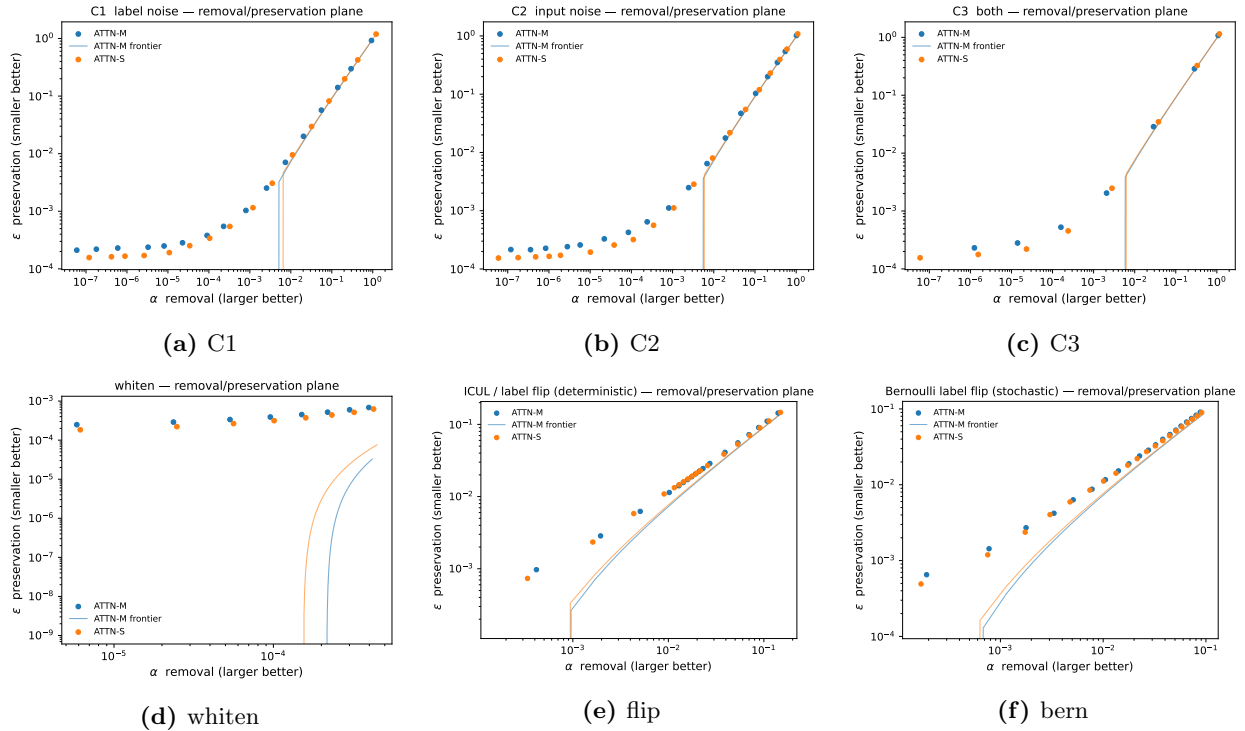


Figure 11: Frontiers, MNIST.

D Remaining tradeoff curves

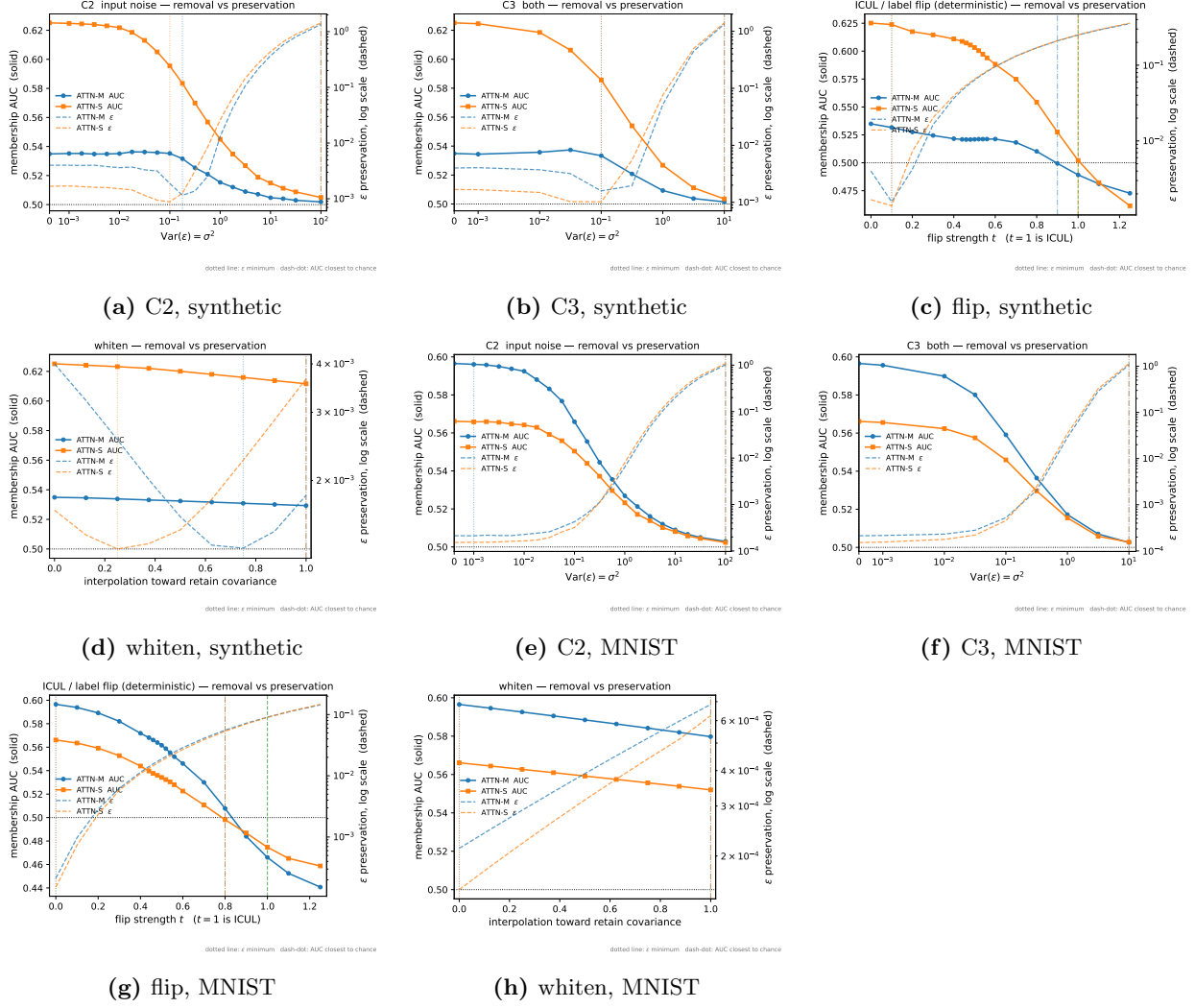


Figure 12: Tradeoff curves for the arms not shown in Figures 4 and 7. AUC solid on the left axis, ϵ dashed on the right.